

Stagehand Phase Evidence

Current checkpoint: public Site v76, verified 2026-09-03.

Project

Stagehand is a WebMCP-native 2D animation studio for editable paper-cutout scenes. The current cut supports rigged characters, imported image props, per-asset visual direction, camera work, captions, non-voice audio cues, storyboard beats, multi-scene Preview, and deterministic WebM export.

Private source: github.com/swaymun/stagehand-animation-studio

Commit: 819b5504e3b94fb67931605260900324d9885f59

Public Site: stagehand-animation-studio.saimun-h-shahee.chatgpt.site

Sites version: 76

Acceptance loop

Implement - local tests - Sites deployment - hosted test - UI roast - fixes - second review.

Implemented at this checkpoint

- Revision-safe WebMCP mutations with optional expectedRevision and structured REVISION_CONFLICT results.
- Idempotent WebMCP mutations with optional idempotencyKey replay protection.
- Structured per-asset role, treatment, silhouette, palette, and direction notes with human and agent parity.
- Motion actions show affected character and bounded duration; pose presets remain a separate group.
- Agent frame-rate-only render updates preserve the existing resolution until a new preset is explicitly chosen.
- Storyboard mode selects the Board rail; Preview hides editing mutation surfaces and selection chrome.

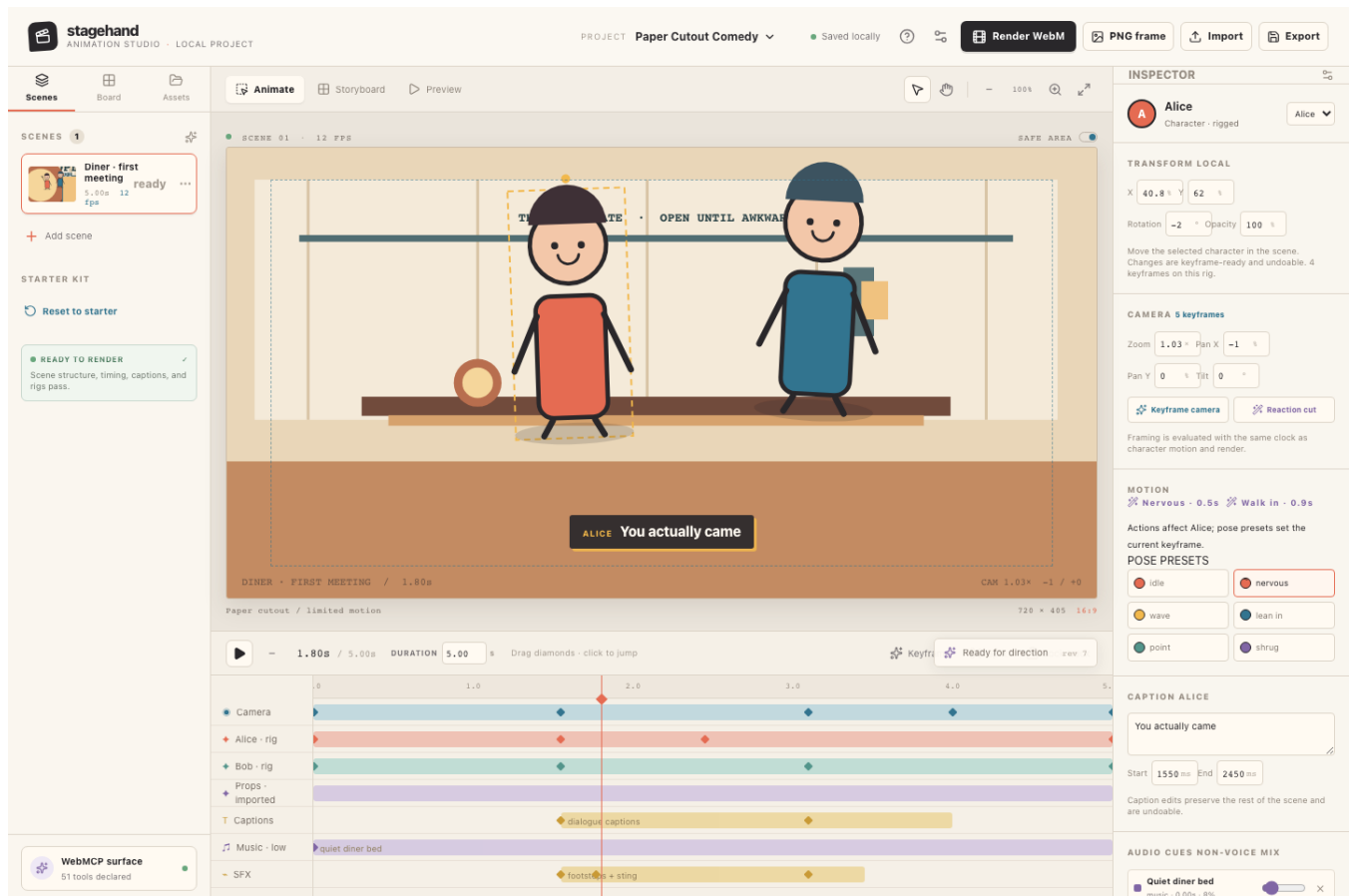
Verification

Local gate:

```
npm run format
npm run lint
npm run build
git diff --check
npm run smoke
```

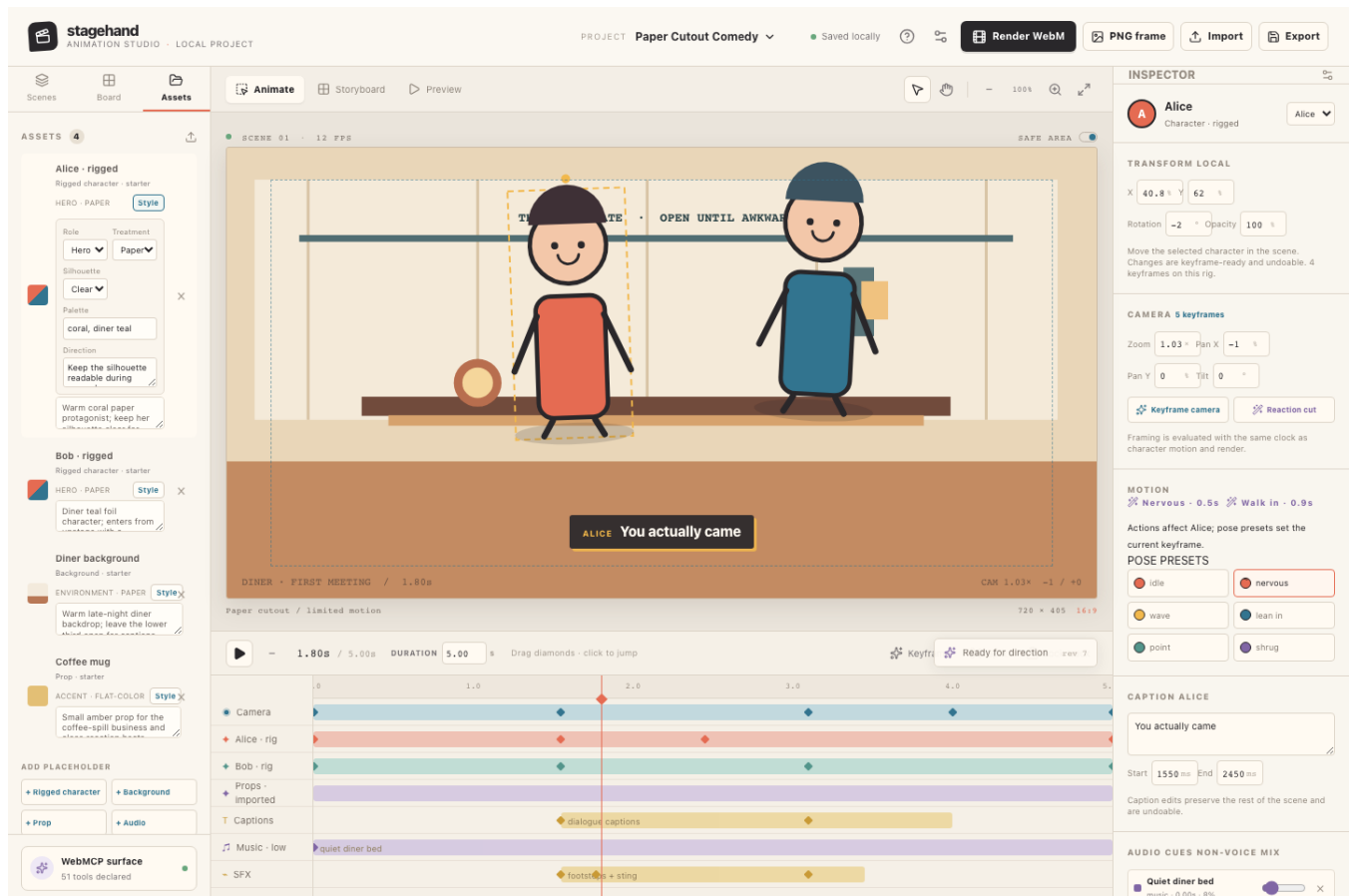
Local and hosted smoke passed: 51 tools, 38 guarded mutations, stale-write conflict, idempotent replay, frame and PNG export, asset-style parity, prop animation, undo/redo, two-scene WebM, sequence Preview transition, Help modal semantics, reload recovery, and zero page errors.

Animate editing workspace



Hosted Playwright capture at 1440 x 960. Motion actions identify their target character and duration; the scene label remains readable.

Assets style editor



Hosted Playwright capture at 1440 x 960. The expandable Style control exposes structured visual direction without taking over the asset rail.

Storyboard mode with Board rail

The screenshot displays the Stagehand Animation Studio interface in storyboard mode. The top bar shows the project name "Paper Cutout Comedy" and various utility buttons like "Render WebM", "PNG frame", "Import", and "Export". The main workspace is divided into three vertical panels: "BEATS", "Storyboard", and "INSPECTOR".

BEATS Panel: Lists three beats: "01 The wait", "02 The entrance", and "03 The pause". Beat 02 is selected and highlighted with a red border.

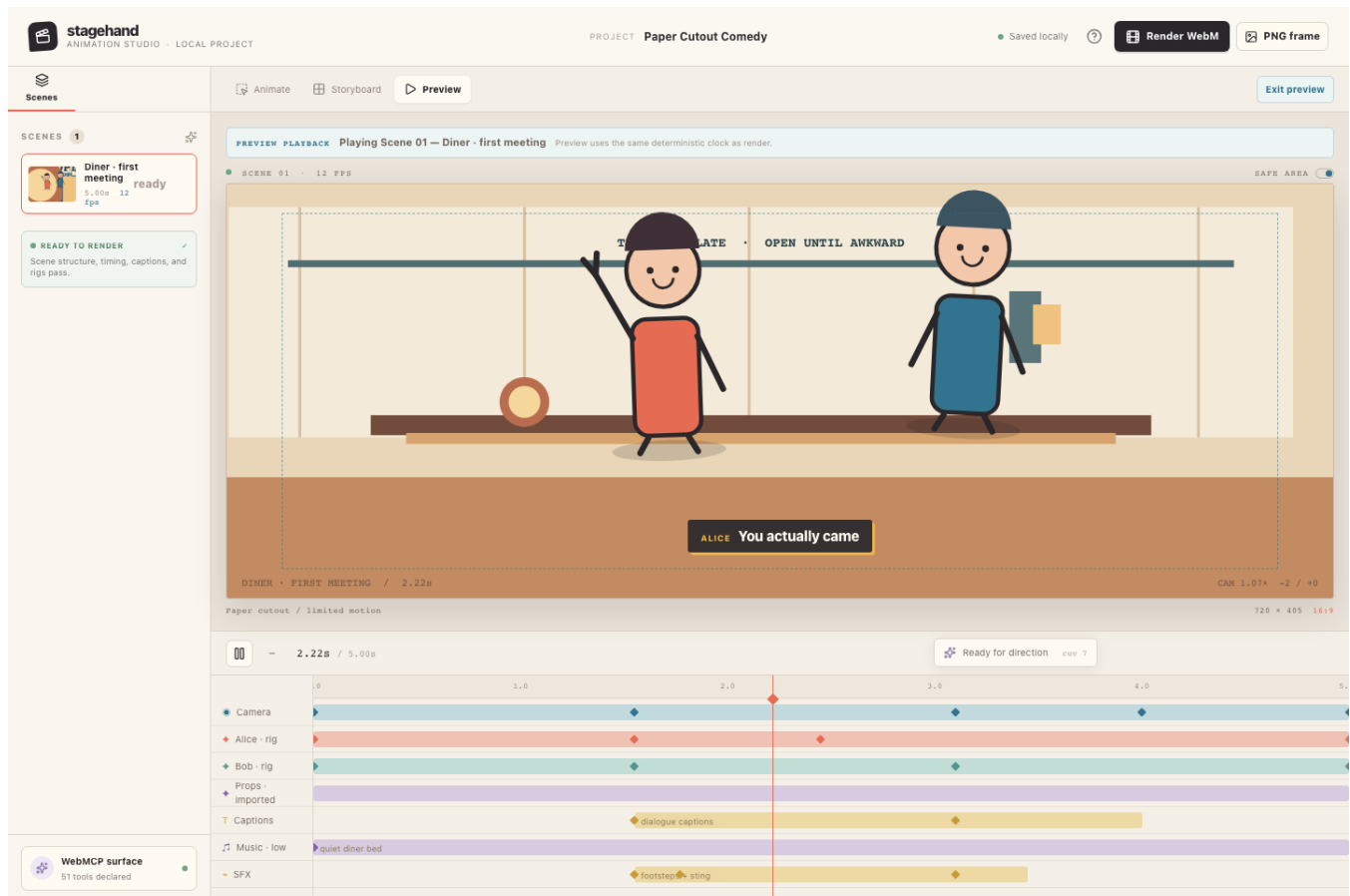
Storyboard Panel: Displays a storyboard titled "Block the awkward moment". It features three panels showing character interactions. The first panel is "The wait" (Alice practices what to say, 0.90s - 1.55s). The second panel is "The entrance" (Bob arrives behind her, 1.55s - 3.10s) and is highlighted with a red border. The third panel is "The pause" (Neither knows what to do, 3.10s - 5.00s). Above the panels is a "BEAT BOARD" with a timeline and a "BOARD RAIL" showing the sequence of beats.

INSPECTOR Panel: Shows the selected character "Alice" and various settings. It includes a "TRANSFORM LOCAL" section for position, rotation, and opacity. A "CAMERA" section shows keyframes for zoom and pan. A "MOTION" section includes "Nervous" and "Walk in" actions. A "POSE PRESETS" section shows "idle" and "nervous" poses. A "CAPTION ALICE" section shows the caption "You actually came" with start and end times (1550ms to 2450ms). An "AUDIO CUES NON-VOICE MIX" section shows a "Quiet diner bed" cue.

The bottom of the interface features a "WebMCP surface" and a "Storyboard" timeline with a playhead at 1.80s. The timeline includes tracks for Camera, Alice - rig, Bob - rig, Props - imported, Captions, Music - low, and SFX.

Hosted Playwright capture at 1440 x 960. The selected top-level mode and project rail stay synchronized.

Review-first Preview player



Hosted Playwright capture at 1440 x 960. Scene-only context, transport, and Exit preview remain; global editing controls are hidden.

Roast findings and fixes

- Imported props lacked direct human transform editing. Added X/Y/Scale/Rot controls sharing the keyframe command path.
- Four prop transform fields initially collided with the asset-list flex rule and truncated. Replaced it with an explicit two-column grid; computed field boxes are 57 px wide in two rows.
- Preview carried too much editing chrome. Converted it into a review-first player with scene-only context and Exit preview.
- Timeline diamonds had 7 x 7 px hit targets. Enlarged the buttons to 22 x 22 pixels while preserving the visual diamond.
- Agent retries could duplicate successful writes. Added optional idempotency-key replay protection and a hosted smoke assertion.
- Switching top-level Storyboard mode could leave Assets selected. Synced mode selection to its companion project rail.
- Character transform inputs lacked explicit accessible names. Added dynamic X/Y/rotation/opacity labels and a smoke check.
- Asset briefs carried style intent only as prose. Added structured role, treatment, silhouette, palette, and direction fields.
- Motion action buttons were visually mixed with pose presets. Added explicit durations, affected-character copy, and a Pose presets label.
- Agent frame-rate-only render updates reset an existing 1080p choice. Preserved the current resolution when the preset is omitted and added smoke coverage.
- Preview still exposed global project-editing controls. Hid Settings, Import, Export JSON, and rename while retaining WebM and PNG review actions.
- Scene titles competed with the ready badge and were visually shortened. Used a two-line title clamp and positioned the badge outside the text flow.
- The smoke gate did not exercise recovery, modal semantics, or cross-scene playback. Added Help open/close, reload recovery, and sequence Preview transition checks to local and hosted runs.

Limits

Direct CUA inspection was unavailable because the Mac was locked; Playwright was used for hosted interaction and screenshots. The smoke harness proves injected modelContext registration and execution, not live production ChatGPT WebMCP discovery. TTS, voice cloning, dialogue recording, phoneme extraction, and lip-sync remain out of scope.